
Lab Report - Exercise 1

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1 Exercise 1

1. This command uses find tool (find /) to scan the entire file system from path "/" (whole file system). type -f: only files not directories are scanned. -exec grep: for every found file, does the command "grep" (grep is a command-line for text searching) -l "password": Only files contain the "password" string is shown.

This command is really slow because it scans the entire system and also may get permission denied.

```
10.79.8.0/28 dev wlan0 proto kernel scope link src 10.79.8.8 metric 600

(root@amir-Kali)-[~/Desktop/LAB1]
# find / -type f -exec grep -l "password" {} \;
find: '/run/user/1001/doc': Permission denied
find: '/run/user/1001/gvfs': Permission denied
/var/cache/man/index.db
/var/cache/man/cs/index.db
/var/cache/man/fr/index.db
/var/cache/man/it/index.db
/var/cache/man/ja/index.db
/var/cache/man/zh_CN/index.db
/var/cache/man/ru/index.db
/var/cache/man/sv/index.db
/var/cache/man/de/index.db
/var/cache/apt/srcpkgcache.bin
/var/cache/apt/pkgcache.bin
/var/cache/debconf/templates.dat
/var/cache/debconf/config.dat-old
/var/cache/debconf/config.dat
/var/cache/debconf/templates.dat-old
/var/cache/cracklib/cracklib_dict.pwd
/var/backups/dpkg.status.0
/var/log/installer/status
/var/log/installer/cdebconf/questions.dat
/var/log/installer/cdebconf/templates.dat
/var/log/installer/syslog
/var/log/journal/233f6815474f40cc923d318cd80f1422/user-1001.journal
/var/log/journal/233f6815474f40cc923d318cd80f1422/user-1000@0a891c6aabe840a5975212cf297dda24-00000000000000556-000650b384caf704.journal
/var/log/journal/233f6815474f40cc923d318cd80f1422/system@0a891c6aabe840a5975212cf297dda24-00000000000000001-000650b383cccbbf.journal
/var/log/journal/233f6815474f40cc923d318cd80f1422/system.journal
/var/log/boot.log.1
/var/lib/dpkg/status
/var/lib/dpkg/status-old
/var/lib/dpkg/available
/var/lib/dpkg/info/libgssapi-krb5-2:amd64.symbols
/var/lib/dpkg/info/libatril-document3t64:amd64.symbols
/var/lib/dpkg/info/libglibmm-2.68-1t64:amd64.symbols
/var/lib/dpkg/info/systemd.md5sums
/var/lib/dpkg/info/sudo.prerm
/var/lib/dpkg/info/kali-themes-common.md5sums
/var/lib/dpkg/info/libqmi-glib5:amd64.symbols
/var/lib/dpkg/info/openssh-server.postinst
/var/lib/dpkg/info/gnome-accessibility-themes.list
^X@ss/var/lib/dpkg/info/libldap2:amd64.symbols
/var/lib/dpkg/info/libqt6network6:amd64.symbols
/var/lib/dpkg/info/atril-common.md5sums
/var/lib/dpkg/info/evolution-data-server-common.list
/var/lib/dpkg/info/gnome-accessibility-themes.md5sums
/var/lib/dpkg/info/kali-menu.list
/var/lib/dpkg/info/libplymouth5:amd64.symbols
/var/lib/dpkg/info/gcr.list
/var/lib/dpkg/info/zenity-common.md5sums
/var/lib/dpkg/info/libgtk-3-0t64:amd64.symbols
/var/lib/dpkg/info/libqt5network5t64:amd64.symbols
/var/lib/dpkg/info/libkrb5-3:amd64.symbols
/var/lib/dpkg/info/plymouth.md5sums
/var/lib/dpkg/info/libpam-runtime.md5sums
/var/lib/dpkg/info/libaccountsservice0:amd64.symbols
```

2. This command first creates a file using printf containing multiple lines and then if it's ok, and then

use sed command (-n by default) which extracts and prints all lines between patterns "start" and "end"(inclusive) and saves them into a new file extracted.txt.

```
(root@amir-Kali)-[~/Desktop/LAB1]
# printf "line 1\nstart\nimportant line 1\nimportant line 2\nend\nlineafter\n" > file.txt && sed -n '/start/,/end/p' file.txt > extracted.txt

(root@amir-Kali)-[~/Desktop/LAB1]
# cat extracted.txt
start
important line 1
important line 2
end

(root@amir-Kali)-[~/Desktop/LAB1]
# cat file.txt
line 1
start
important line 1
important line 2
end
lineafter
```

3. This command uses awk to process the /etc/passwd files. It sets the field separator to ":" and prints only users with UID greater or equal to 1000, which typically represents regular users. The Output includes the username(\$1), UID(\$3), and home director(\$6).

```
(root@amir-Kali)-[~/Desktop/LAB1]
# awk 'BEGIN{FS=":"; OFS=":"} {if ($3 >= 1000) print $1, $3, $6}' /etc/passwd
wd
nobody:65534:/nonexistent
kalikali:1000:/home/kalikali
basic:1001:/home/basic

(root@amir-Kali)-[~/Desktop/LAB1]
# cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailng List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
tss:x:990:990:tss user for tpm2:/:/usr/sbin/nologin
strongswan:x:100:65534::/var/lib/strongswan:/usr/sbin/nologin
systemd-timesync:x:989:989:systemd Time Synchronization:/:/usr/sbin/nologin
messagebus:x:988:988:System Message Bus:/nonexistent:/usr/sbin/nologin
tcpdump:x:987:987:tcpdump:/nonexistent:/usr/sbin/nologin
_rpc:x:101:65534::/run/rpcbind:/usr/sbin/nologin
statd:x:102:65534::/var/lib/nfs:/usr/sbin/nologin
sshd:x:985:65534:sshd user:/run/ssh:/usr/sbin/nologin
dnsmasq:x:984:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:103:105:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
speech-dispatcher:x:983:983:Speech Dispatcher:/run/speech-dispatcher:/bin/false
usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
nm-openvpn:x:982:982:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
pipewire:x:981:981:system user for pipewire:/nonexistent:/usr/sbin/nologin
nm-openconnect:x:980:980:NetworkManager OpenConnect plugin:/:/usr/sbin/nologin
saned:x:105:106::/var/lib/saned:/usr/sbin/nologin
lightdm:x:106:107:Light Display Manager:/var/lib/lightdm:/bin/false
polkitd:x:979:979:User for polkitd:/:/usr/sbin/nologin
rtkit:x:978:978:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:977:977:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
pcscd:x:976:976:PC/SC Smart Card Daemon:/:/usr/sbin/nologin
kalikali:x:1000:1000:Kalikali,,,:/home/kalikali:/usr/bin/zsh
basic:x:1001:1001::/home/basic:/bin/bash
```

4. This command executes two commands in a subshell in (ls /etc and cat nonexistentfile in paranthesis so it's going to be executed in group). The output of ls /etc os redirected to stdout.log, while the error generated by "cat nonexistentfile" is redirected to stderr.log.

```
Session Actions Edit View Help

(root@amir-Kali) [~/Desktop/LAB1]
# (ls /etc; cat nonexistentfile) > stdout.log 2> stderr.log

(root@amir-Kali) [~/Desktop/LAB1]
# cat stderr.log
cat: nonexistentfile: No such file or directory

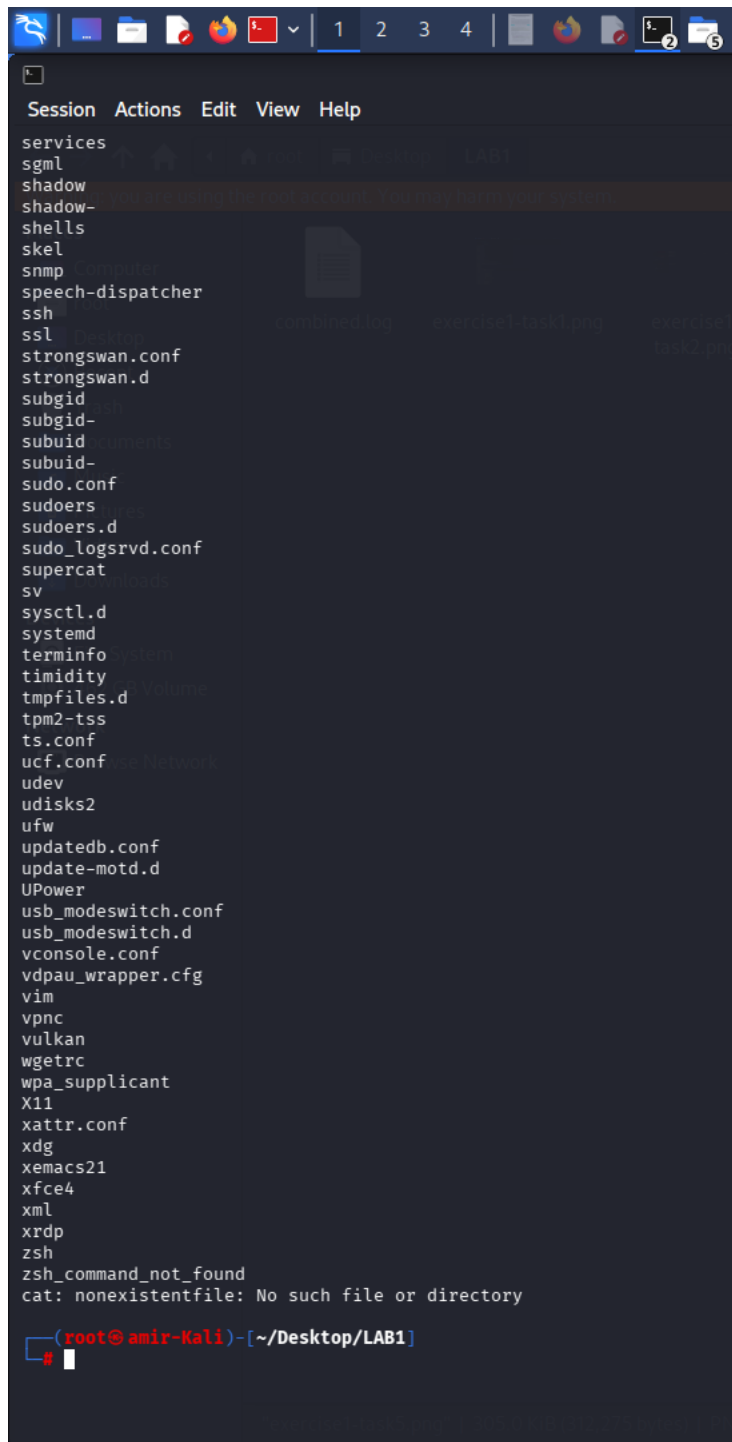
(root@amir-Kali) [~/Desktop/LAB1]
# cat stdout.log
adduser.conf
adjtime
alternatives
apparmor
apparmor.d
apt
avahi
bash.bashrc
bash_completion
bash_completion.d
bindresvport.blacklist
binfmt.d
bluetooth
ca-certificates
ca-certificates.conf
chatscripts
chromium
cloud
colord
console-setup
cracklib
credstore
credstore.encrypted
cron.d
cron.daily
cron.hourly
cron.monthly
crontab
cron.weekly
cron.yearly
dbus-1
dconf
debconf.conf
debian_version
default
deluser.conf
depmod.d
dhcpcd
dhcpcd.conf
dictionaries-common
dpkg
e2scrub.conf
eac
emacs
environment
environment.d
ethertypes
firefox-esr
fonts
fstab
fuse.conf
```

5. The same as previous task, but the different thing is first of all the output of first statement is returned in "combined.log" file and also, the second statement is merged at the end of it in the same file (because of &1).



A terminal window titled "Session Actions Edit View Help" is shown. The prompt is `(root@amir-Kali)-[~/Desktop/LAB1]`. The user enters the command `# (ls /etc; cat nonexistentfile) > combined.log 2>&1`. The prompt changes to `(root@amir-Kali)-[~/Desktop/LAB1]` and the user enters `# cat combined.log`. The output is a list of files in `/etc`:

```
adduser.conf
adjtime
alternatives
apparmor
apparmor.d
apt
avahi
bash.bashrc
bash_completion
bash_completion.d
bindresvport.blacklist
binfmt.d
bluetooth
ca-certificates
ca-certificates.conf
chatscripts
chromium
cloud
colord
colord
console-setup
cracklib
credstore
credstore.encrypted
cron.d
cron.daily
cron.hourly
cron.monthly
crontab
cron.weekly
cron.yearly
dbus-1
dconf
debconf.conf
debian_version
default
deluser.conf
depmod.d
dhcp
dhcpcd.conf
dictionaries-common
dpkg
e2scrub.conf
eac
emacs
environment
environment.d
ethertypes
firefox-esr
fonts
fstab
fuse.conf
gai.conf
ghostscript
glvnd
gnome-system-tools
groff
```



2 Exercise 2

2.1 a)

First of all for finding active machines in subnet, we use flag -sn (means Null scan and doesn't set any bits or TCP flag header is 0) in nmap which only uses ARP requests to find them.

```

(root@amir-Kali)-[~]
# nmap -sn 10.10.16.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-03 16:07 +0200
Nmap scan report for 10.10.16.1
Host is up (0.028s latency).
Nmap scan report for 10.10.16.9
Host is up (0.034s latency).
Nmap scan report for 10.10.16.47
Host is up (0.030s latency).
Nmap scan report for 10.10.16.54
Host is up (0.029s latency).
Nmap scan report for 10.10.16.55
Host is up (0.028s latency).
Nmap scan report for 10.10.16.65
Host is up (0.028s latency).
Nmap scan report for 10.10.16.77
Host is up (0.031s latency).
Nmap scan report for 10.10.16.82
Host is up (0.031s latency).
Nmap scan report for 10.10.16.87
Host is up (0.027s latency).
Nmap scan report for 10.10.16.89
Host is up (0.027s latency).
Nmap scan report for 10.10.16.90
Host is up (0.027s latency).
Nmap scan report for 10.10.16.99
Host is up (0.033s latency).
Nmap scan report for 10.10.16.101
Host is up (0.033s latency).
Nmap scan report for 10.10.16.102
Host is up (0.034s latency).
Nmap scan report for 10.10.16.147
Host is up (0.030s latency).
Nmap scan report for 10.10.16.154
Host is up (0.027s latency).
Nmap scan report for 10.10.16.166
Host is up (0.031s latency).
Nmap scan report for 10.10.16.203
Host is up (0.024s latency).
Nmap scan report for 10.10.16.237
Host is up (0.028s latency).
Nmap scan report for 10.10.16.239
Host is up (0.024s latency).
Nmap scan report for 10.10.16.240
Host is up (0.024s latency).
Nmap scan report for 10.10.16.241
Host is up (0.024s latency).
Nmap scan report for 10.10.16.242
Host is up (0.024s latency).
Nmap scan report for 10.10.16.246
Host is up (0.028s latency).
Nmap done: 256 IP addresses (24 hosts up) scanned in 5.86 seconds

```

Then, in order to do full port scan, we use `sudo nmap -sS -p- -T4 -oN fullportscan.txt 10.10.16.0/24`. we use `sudo -sS` which means SYN scan and with root privileges for quicker scan. `-p-` indicates all ports (1-65535); `-T4`: Increases speed for faster execution; also we write the output into file `fullportscan.txt`. P.N: As you can see in both parts of a) we have 24 hosts up. In following text, you can see the final output of part 1:

```

1 # Nmap 7.99 scan initiated Sun May 3 17:22:34 2026 as: /usr/lib/nmap/nmap -sS -p- -T4 -oN
   fullportscan.txt 10.10.16.0/24
2 Nmap scan report for 10.10.16.1
3 Host is up (0.027s latency).
4 Not shown: 65531 closed tcp ports (reset)
5 PORT      STATE SERVICE
6 53/tcp    open  domain
7 2222/tcp   open  EtherNetIP-1
8 5000/tcp   open  upnp
9 9000/tcp   open  cslistener
10
11 Nmap scan report for 10.10.16.9
12 Host is up (0.028s latency).
13 Not shown: 65532 closed tcp ports (reset)
14 PORT      STATE SERVICE
15 22/tcp    open  ssh
16 139/tcp   open  netbios-ssn
17 445/tcp   open  microsoft-ds
18
19 Nmap scan report for 10.10.16.36
20 Host is up (0.027s latency).
21 Not shown: 63397 closed tcp ports (reset), 2137 filtered tcp ports (no-response)
22 PORT      STATE SERVICE
23 22/tcp    open  ssh
24
25 Nmap scan report for 10.10.16.47
26 Host is up (0.029s latency).
27 Not shown: 65533 closed tcp ports (reset)
28 PORT      STATE SERVICE

```

```

29 22/tcp open  ssh
30 80/tcp open  http
31
32 Nmap scan report for 10.10.16.54
33 Host is up (0.032s latency).
34 Not shown: 65533 closed tcp ports (reset)
35 PORT      STATE SERVICE
36 21/tcp    open  ftp
37 22/tcp    open  ssh
38
39 Nmap scan report for 10.10.16.55
40 Host is up (0.025s latency).
41 Not shown: 65533 closed tcp ports (reset)
42 PORT      STATE SERVICE
43 22/tcp    open  ssh
44 80/tcp    open  http
45
46 Nmap scan report for 10.10.16.65
47 Host is up (0.028s latency).
48 Not shown: 65534 closed tcp ports (reset)
49 PORT      STATE SERVICE
50 22/tcp    open  ssh
51
52 Nmap scan report for 10.10.16.77
53 Host is up (0.030s latency).
54 Not shown: 65533 closed tcp ports (reset)
55 PORT      STATE SERVICE
56 22/tcp    open  ssh
57 80/tcp    open  http
58
59 Nmap scan report for 10.10.16.82
60 Host is up (0.037s latency).
61 Not shown: 65533 closed tcp ports (reset)
62 PORT      STATE SERVICE
63 22/tcp    open  ssh
64 10100/tcp open  itap-ddtp
65
66 Nmap scan report for 10.10.16.87
67 Host is up (0.033s latency).
68 Not shown: 65533 closed tcp ports (reset)
69 PORT      STATE SERVICE
70 22/tcp    open  ssh
71 80/tcp    open  http
72
73 Nmap scan report for 10.10.16.89
74 Host is up (0.025s latency).
75 Not shown: 65532 filtered tcp ports (no-response)
76 PORT      STATE SERVICE
77 22/tcp    open  ssh
78 80/tcp    open  http
79 9090/tcp  open  zeus-admin
80
81 Nmap scan report for 10.10.16.90
82 Host is up (0.032s latency).
83 Not shown: 65531 closed tcp ports (reset)
84 PORT      STATE SERVICE
85 21/tcp    open  ftp
86 22/tcp    open  ssh
87 80/tcp    open  http
88 8080/tcp  open  http-proxy
89
90 Nmap scan report for 10.10.16.99
91 Host is up (0.037s latency).
92 Not shown: 65534 closed tcp ports (reset)
93 PORT      STATE SERVICE
94 22/tcp    open  ssh
95
96 Nmap scan report for 10.10.16.101
97 Host is up (0.032s latency).
98 Not shown: 65534 closed tcp ports (reset)
99 PORT      STATE SERVICE

```



```

100 22/tcp open  ssh
101
102 Nmap scan report for 10.10.16.102
103 Host is up (0.039s latency).
104 Not shown: 65534 closed tcp ports (reset)
105 PORT      STATE SERVICE
106 22/tcp open  ssh
107
108 Nmap scan report for 10.10.16.147
109 Host is up (0.034s latency).
110 Not shown: 65534 closed tcp ports (reset)
111 PORT      STATE SERVICE
112 22/tcp open  ssh
113
114 Nmap scan report for 10.10.16.154
115 Host is up (0.035s latency).
116 Not shown: 65534 closed tcp ports (reset)
117 PORT      STATE SERVICE
118 22/tcp open  ssh
119
120 Nmap scan report for 10.10.16.203
121 Host is up (0.030s latency).
122 Not shown: 65534 closed tcp ports (reset)
123 PORT      STATE SERVICE
124 22/tcp open  ssh
125
126 Nmap scan report for 10.10.16.237
127 Host is up (0.030s latency).
128 Not shown: 65534 closed tcp ports (reset)
129 PORT      STATE SERVICE
130 22/tcp open  ssh
131
132 Nmap scan report for 10.10.16.239
133 Host is up (0.036s latency).
134 Not shown: 65534 closed tcp ports (reset)
135 PORT      STATE SERVICE
136 22/tcp open  ssh
137
138 Nmap scan report for 10.10.16.240
139 Host is up (0.030s latency).
140 Not shown: 65534 closed tcp ports (reset)
141 PORT      STATE SERVICE
142 22/tcp open  ssh
143
144 Nmap scan report for 10.10.16.241
145 Host is up (0.030s latency).
146 Not shown: 65534 closed tcp ports (reset)
147 PORT      STATE SERVICE
148 22/tcp open  ssh
149
150 Nmap scan report for 10.10.16.242
151 Host is up (0.031s latency).
152 Not shown: 65534 closed tcp ports (reset)
153 PORT      STATE SERVICE
154 22/tcp open  ssh
155
156 Nmap scan report for 10.10.16.246
157 Host is up (0.031s latency).
158 Not shown: 65533 closed tcp ports (reset)
159 PORT      STATE SERVICE
160 22/tcp open  ssh
161 1337/tcp open waste
162
163 # Nmap done at Sun May  3 17:40:36 2026 -- 256 IP addresses (24 hosts up) scanned in 1081.67
    seconds

```

Listing 1: Full port scan results

2.2 b)

First, I scanned the subnet 10.10.0.0/19 to find hosts using `nmap -sn 10.10.0.0/19`. After that I selected one, e.g. 10.10.11.9. After that, I opened Wireshark and scanned the host with different nmap scan flags. The following listing shows the outputs of a default nmap scan, `nmap -sS` for a SYN scan, `nmap -sV` to detect services versions and `nmap -A` to detect OS, Scripts and Versions.

```
1 nmap 10.10.11.9
2 Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-07 01:16 -0400
3 Nmap scan report for 10.10.11.9
4 Host is up (0.032s latency).
5 Not shown: 997 closed tcp ports (reset)
6 PORT      STATE SERVICE
7 22/tcp    open  ssh
8 139/tcp   open  netbios-ssn
9 445/tcp   open  microsoft-ds
10
11 Nmap done: 1 IP address (1 host up) scanned in 1.25 seconds
12
13
14
15 nmap -sS 10.10.11.9
16 Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-07 01:18 -0400
17 Nmap scan report for 10.10.11.9
18 Host is up (0.032s latency).
19 Not shown: 997 closed tcp ports (reset)
20 PORT      STATE SERVICE
21 22/tcp    open  ssh
22 139/tcp   open  netbios-ssn
23 445/tcp   open  microsoft-ds
24
25 Nmap done: 1 IP address (1 host up) scanned in 1.28 seconds
26
27
28
29
30 nmap -sV 10.10.11.9
31 Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-07 01:18 -0400
32 Nmap scan report for 10.10.11.9
33 Host is up (0.040s latency).
34 Not shown: 997 closed tcp ports (reset)
35 PORT      STATE SERVICE      VERSION
36 22/tcp    open  ssh          OpenSSH 8.9p1 Ubuntu 3ubuntu0.11 (Ubuntu Linux; protocol 2.0)
37 139/tcp   open  netbios-ssn Samba smbd 4
38 445/tcp   open  netbios-ssn Samba smbd 4
39 Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
40
41 Service detection performed. Please report any incorrect results at https://nmap.org/submit/
42 Nmap done: 1 IP address (1 host up) scanned in 12.96 seconds
43
44
45
46
47 nmap -A 10.10.11.9
48 Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-07 01:18 -0400
49 Nmap scan report for 10.10.11.9
50 Host is up (0.033s latency).
51 Not shown: 997 closed tcp ports (reset)
52 PORT      STATE SERVICE      VERSION
53 22/tcp    open  ssh          OpenSSH 8.9p1 Ubuntu 3ubuntu0.11 (Ubuntu Linux; protocol 2.0)
54 | ssh-hostkey:
55 |   256 75:c8:93:8e:a0:3c:d2:ac:ce:5b:e6:bc:0b:83:76:2e (ECDSA)
56 |_  256 43:e0:74:c4:8c:7f:21:67:a2:bc:22:5d:7c:b1:ef:c8 (ED25519)
57 139/tcp   open  netbios-ssn Samba smbd 4
58 445/tcp   open  netbios-ssn Samba smbd 4
59 No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
60 TCP/IP fingerprint:
61 OS: SCAN(V=7.98%E=4%D=5/7%OT=22%CT=1%CU=43150%PV=Y%DS=2%DC=T%G=Y%TM=69FC20DC
62 OS:%P=x86_64-pc-linux-gnu)SEQ(SP=105%GCD=1%ISR=10A%TI=Z%CI=Z%II=I%TS=A)SEQ(
```

```

63 OS:SP=106%GCD=1%ISR=10B%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=106%GCD=1%ISR=10E%TI=Z%C
64 OS:I=Z%II=I%TS=A)SEQ(SP=FF%GCD=1%ISR=105%TI=Z%CI=Z%II=I%TS=A)OPS(O1=M45OST1
65 OS:1NW6%O2=M45OST11NW6%O3=M45ONNT11NW6%O4=M45OST11NW6%O5=M45OST11NW6%O6=M45
66 OS:OST11)WIN(W1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T
67 OS:=40%W=FAFO%O=M45ONNSNW6%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=AS%RD=O%Q=)T
68 OS:2(R=N)T3(R=N)T4(R=Y%DF=Y%T=40%W=O%S=A%A=Z%F=R%O=%RD=O%Q=)T5(R=Y%DF=Y%T=4
69 OS:O%W=O%S=Z%A=S+%F=AR%O=%RD=O%Q=)T6(R=Y%DF=Y%T=40%W=O%S=A%A=Z%F=R%O=%RD=O%
70 OS:Q=)T7(R=Y%DF=Y%T=40%W=O%S=Z%A=S+%F=AR%O=%RD=O%Q=)U1(R=Y%DF=N%T=40%IPL=16
71 OS:4%UN=O%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)
72
73 Network Distance: 2 hops
74 Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
75
76 Host script results:
77 | smb2-security-mode:
78 |   3.1.1:
79 |_ Message signing enabled but not required
80 | smb2-time:
81 |   date: 2026-05-07T05:19:21
82 |_ start_date: N/A
83 |_nbstat: NetBIOS name: UBUNTU, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
84 |_clock-skew: -2s
85
86 TRACEROUTE (using port 1723/tcp)
87 HOP RTT ADDRESS
88 1 45.04 ms 10.8.0.1
89 2 40.17 ms 10.10.11.9
90
91 OS and Service detection performed. Please report any incorrect results at https://nmap.org/
submit/ .
92 Nmap done: 1 IP address (1 host up) scanned in 26.79 seconds

```

In Wireshark, I filtered for `ip.dst == 10.10.11.9`. Then we compared a SYN Packet to a TCP packet with `nmap -sS` and `nmap -sT`. There we can see, that the SYN flag bit is set to 1 for a SYN packet and the ACK flag bit is set to 1 in a TCP packet.

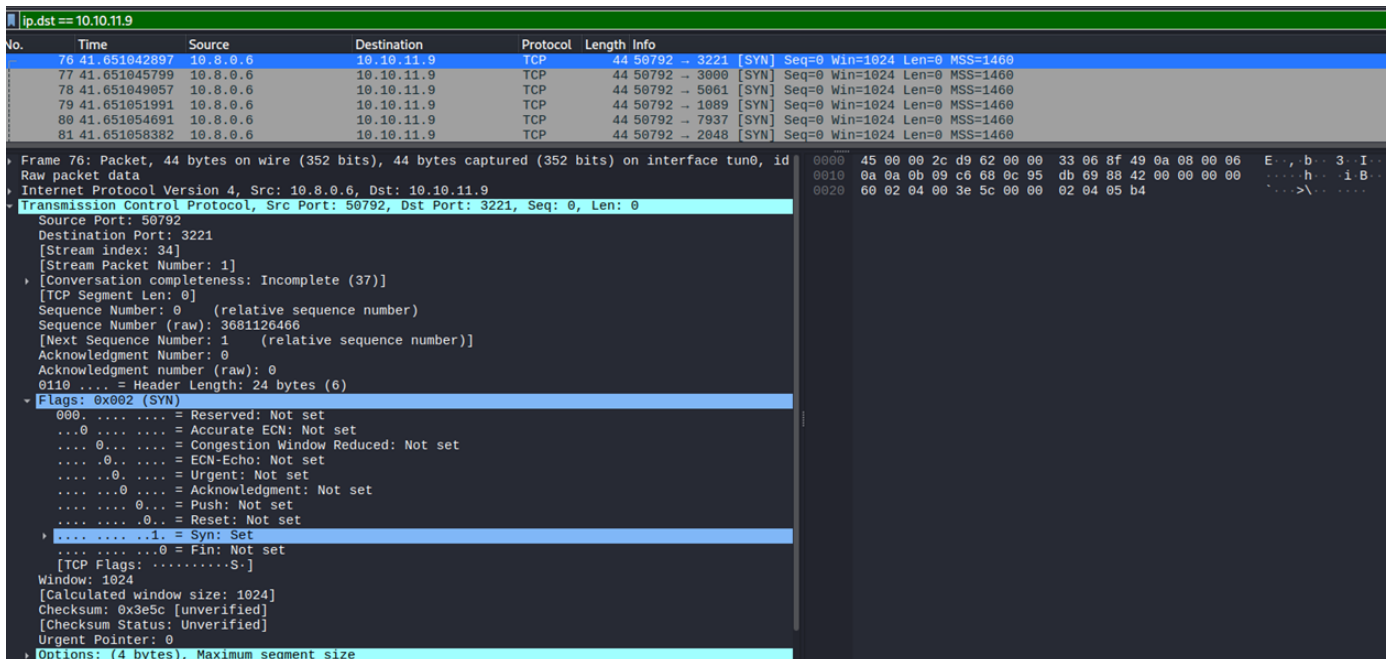


Figure 1: Wireshark screenshot of `nmap -sS` with `SYN=1` and `ACK=0`

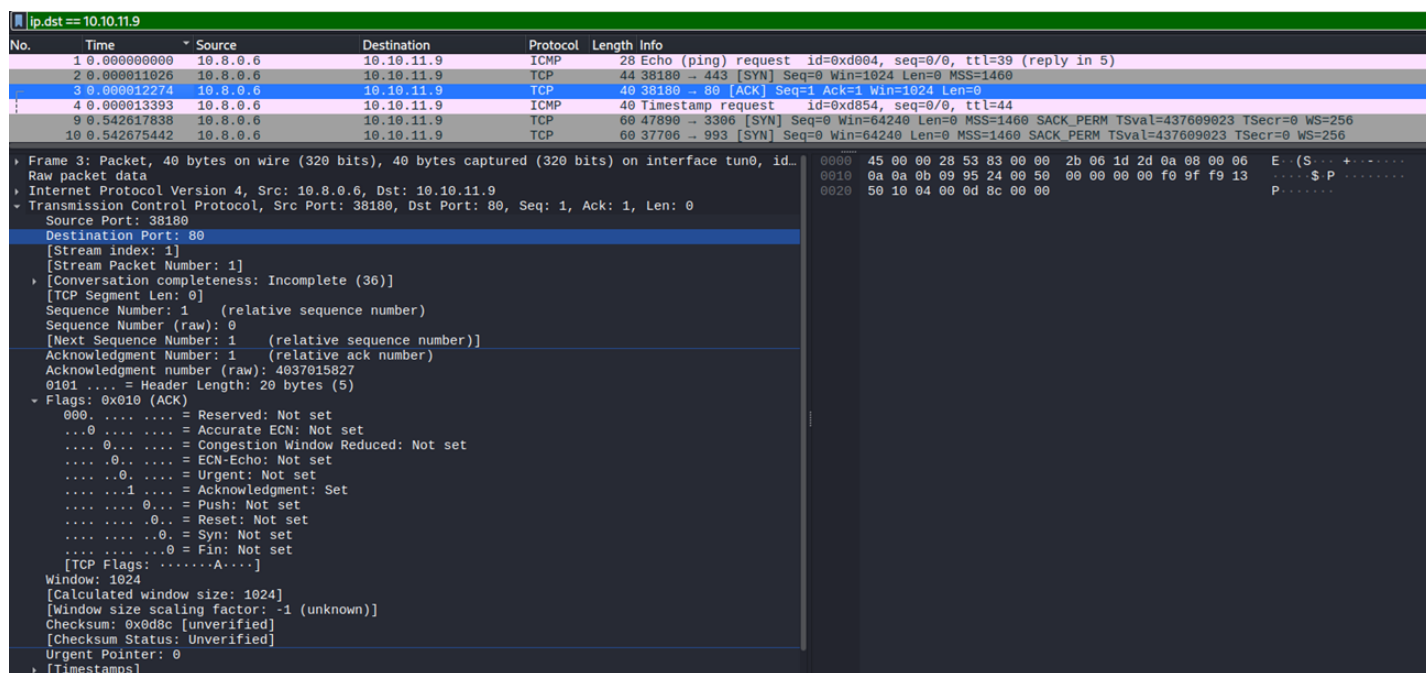


Figure 2: Wireshark screenshot of nmap -sT with SYN=0 and ACK=1

3 Exercise 3

3.1 Identifying Subdirectories

I did the brute force search for directories given wordlist appending each word in the wordlist to the end of IP address of 10.10.16.87. Also I tried out instead of only getting response code 200 for http requests, use 301, 302 for redirection and 403 for forbidden which totally means there is something but not OK response. So there is the result for 403: `http://10.10.16.87/server-status` : which appears as a forbidden page with OS version (Apache/2.4.52 (Ubuntu) Server at 10.10.16.87 Port 80)

```
1 import requests
2
3 target = "http://10.10.16.87"
4 wordlist = "../../Downloads/lab1/wordlist"
5
6 with open(wordlist, "r") as f:
7     for word in f:
8         word = word.strip()
9         url = f"{target}/{word}"
10        try:
11            r = requests.get(url)
12            if r.status_code in [200, 301, 302, 403]:
13                print(f"Found directory: {url}")
14        except:
15            pass
```

```
(root@amir-Kali)-[~/Desktop/LAB1]
# python ./exerc3.py
Found directory: http://10.10.16.87/test
Found directory: http://10.10.16.87/sources
Found directory: http://10.10.16.87/
Found directory: http://10.10.16.87/inprogress
Found directory: http://10.10.16.87/
```

3.2 Searching for files

This is the code I used for searching files in root, and subdirectories. First, I used basic single thread program but because size of the word list is too large, it takes a lot of time. I was wondering how can I improve it with multithreading. In this way, I found out it's better and also correct results.

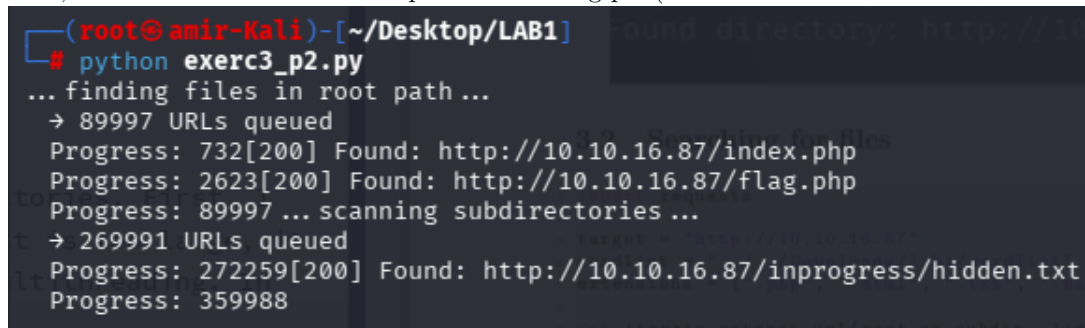
```
1 import requests
2 from concurrent.futures import ThreadPoolExecutor, as_completed
3 from threading import Lock
4 lock = Lock()
5 count = 0
6
7 target = "http://10.10.16.87"
8 wordlist = "../../Downloads/lab1/wordlist"
9 extensions = [".php", ".html", ".txt"]
10 MAX_THREADS = 20
11
12 def send_req(url):
13     global count
14     try:
15         r = requests.get(url, timeout=5)
16         if r.status_code in [200, 301, 302, 403]:
17             print(f"[{r.status_code}] Found: {url}")
18     except:
19         pass
20     with lock:
21         count += 1
22         print(f"\r Progress: {count}", end="", flush=True)
23
24 def build_urls_root(extensions):
25     urls = []
26     with open(wordlist, "r") as f:
27         for word in f:
28             word = word.strip()
29             if not word:
30                 continue
31             for ext in extensions:
32                 urls.append(f"{target}/{word}{ext}")
33     return urls
34
35 def build_urls_subdirs(subdirs):
36     urls = []
37     with open(wordlist, "r") as f:
38         words = [w.strip() for w in f if w.strip()]
39     for subdir in subdirs:
40         for word in words:
41             for ext in extensions:
42                 urls.append(f"{target}/{subdir}/{word}{ext}")
43     return urls
44
45 def run_threaded(urls):
46     with ThreadPoolExecutor(max_workers=MAX_THREADS) as executor:
47         futures = {executor.submit(send_req, url): url for url in urls}
48         for future in as_completed(futures):
49             pass # results already printed inside send_req
50
```

```

51 def files_in_root():
52     print("...finding files in root path...")
53     urls = build_urls_root(extensions)
54     print(f"    -> {len(urls)} URLs queued")
55     run_threaded(urls)
56
57 def scan_subdirs():
58     subdirs = ["test", "sources", "inprogress"]
59     print("...scanning subdirectories...")
60     urls = build_urls_subdirs(subdirs)
61     print(f"    -> {len(urls)} URLs queued")
62     run_threaded(urls)
63
64 def main():
65     files_in_root()
66     scan_subdirs()
67
68 if __name__ == "__main__":
69     main()

```

Also, we can see the result of this part as following pic (There are no other files found in this IP):



```

(root@amir-Kali)-[~/Desktop/LAB1]
# python exerc3_p2.py
... finding files in root path...
→ 89997 URLs queued
Progress: 732[200] Found: http://10.10.16.87/index.php
Progress: 2623[200] Found: http://10.10.16.87/flag.php
Progress: 89997 ... scanning subdirectories ...
→ 269991 URLs queued
Progress: 272259[200] Found: http://10.10.16.87/inprogress/hidden.txt
Progress: 359988

```

With the tool gobuster we could scan for files with
gobuster dir -u http://10.10.16.87 -w wordlist -x php,html,txt

```

kali@kali: ~/Downloads/group5
Session Actions Edit View Help
kali@kali: ~/Downloads/group5
kali@kali: ~/Downloads/group5
(kali@kali)-[~/Downloads/group5]
$ gobuster dir -u http://10.10.16.87 -w /usr/share/wordlists/dirb/common.txt -x php,html,txt

Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://10.10.16.87
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8.2
[+] Extensions: php,html,txt
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

.hta (Status: 403) [Size: 276]
.hta.txt (Status: 403) [Size: 276]
.hta.php (Status: 403) [Size: 276]
.hta.html (Status: 403) [Size: 276]
.htaccess (Status: 403) [Size: 276]
.htaccess.php (Status: 403) [Size: 276]
.htaccess.html (Status: 403) [Size: 276]
.htaccess.txt (Status: 403) [Size: 276]
.htpasswd (Status: 403) [Size: 276]
.htpasswd.php (Status: 403) [Size: 276]
.htpasswd.html (Status: 403) [Size: 276]
.htpasswd.txt (Status: 403) [Size: 276]
flag.php (Status: 200) [Size: 159]
index.php (Status: 200) [Size: 1270]
index.php (Status: 200) [Size: 1270]
server-status (Status: 403) [Size: 276]
sources (Status: 301) [Size: 312] [→ http://10.10.16.87/sources/]
test (Status: 301) [Size: 309] [→ http://10.10.16.87/test/]
Progress: 18452 / 18452 (100.00%)

Finished

(kali@kali)-[~/Downloads/group5]
$ ss

```

4 Exercise 4

4.1 10.10.16.9

`nmap -sV -A 10.10.16.9`: The host 10.10.16.9 exposes SSH (22) and SMB services (139/445). `smbclient -L //10.10.16.9` gives us sambashares where we can access with `smbclient //10.10.16.9/sambashare`. With "ls" and "get flag.txt" we can read the flag{Tm90ZSB0byBzZWxmOiBEb250IGNvbmZpZ3VyZSBzYW1iYSBsaWtlIHRoaXMh}.

4.2 10.10.16.54

After `nmap -sV -A 10.10.16.54` there FTP server with anonymous login allowed online. Login data ist USER and PASS but to access it we need to type "anonymous" and just enter when password is asked. flag{Q2FuIFlvdSBzZWfkiHRoaXM/}

4.3 10.10.16.65

I tried ssh with 10.10.16.65 with user@10.10.16.65 and guessed some password and gained access and then ls and the flag is:

flag{V2VhayBzc2ggaXMgdGhlIGRvb20gb2YgY2l2aWxpc2F0aW9uIQ==}

4.4 10.10.16.87

There is a Webpage found with nmap -sV -A 10.10.16.87.

`gobuster dir -u http://10.10.16.87 -w /usr/share/wordlists/dirb/common.txt -x php,html,txt` gives us several links. On the site /flag there is a flag hidden in the html body flag{MiBGbGFncyAxIFdlYnNpdGU=}.

Hidden in a subdirectory: When I was scanning in exercise 2, I found inprogress directory, and I opened in browser and found a text file named hidden.txt (in link `http://10.10.16.87/inprogress/hidden.txt`); I opened it and found the flag:

flag{Q291bGQgeW91IGZpbmQgdGhpcyBvbWUgYXNMgd2VsbD8=}

The Last flag can be found on `https://10.10.16.87`. After opening inspectors mode and going to storage, the cookie value for the login can be set to true and the site can be refreshed.

flag{Q29va2llcyBhcmUgaW1wb3J0YW50LCBidXQgb2Z0ZW4gd2VhayE=}